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National University of Computer and Emerging Sciences

Feedback Control Systems (EE3004)

Final Exam

Date: June 3rd 2026

Total Time: 3 Hours

Course Instructor

Total Marks: 100

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Total Questions: 05

Semester: *SP-2026*

Campus: Lahore

Dept: Electrical Engg.

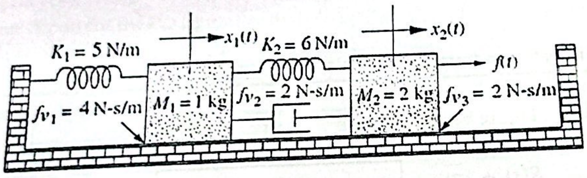
Student Name ____________ Roll No ______ Section ______ *Student Signature* ____________

Vetted by ____________ *Vetter Signature* ____________

CLO # 1: Develop mathematical models of physical systems using *transfer functions* and *state-space methods.*

Q1: ***[20 marks]***

a. **Develop** the state space representation of the following mechanical system. The variable $x_1(t)$ is the output.



b. **Formulate** transfer function of the form, Y(s)/U(s), using the following state space representation. Hence, find the expression of output, $y(t)$, if the input signal, $u(t)$, is a unit-step signal.

$$\dot{x}(t) = \begin{bmatrix} 0 & 2 \\ -2 & -5 \end{bmatrix} x(t) + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 3 & 5 \end{bmatrix} x(t) + [0]\, u(t)$$

$((sI - A)^{-1} B$

CLO # 2: Analyze the transient and steady state response of a feedback system.

Q2: The system shown below has its transient response altered by adding a tachometer. **[20 marks]**

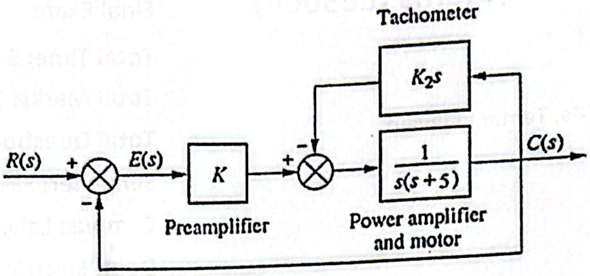
a. Reduce the block diagram to a unity feedback system and **find** the system's open-loop transfer function G(s).

b. Hence, reduce the unity feedback system in (a) to **find** the closed-loop transfer function T(s).

c. Design K and K_2 in the system to yield a damping ratio of **0.80** and a natural frequency of **12.0 rad/s.**

d. Use G(s) in (a) to find the steady-state error of this system for a unit ramp input.

e. What modification is needed in the pre-amplifier K to ensure the steady-state error becomes zero.



CLO # 3: Analyze the stability of a system using analytical methods.

Q3: **[20 marks]**

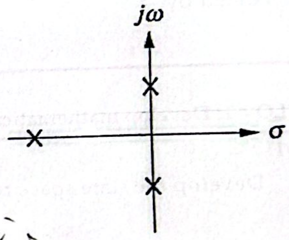
a. For the unity feedback system shown below, where:

$$G(s) = \frac{K}{(s+10)(s^2+4s+5)}$$

i. **Find** the closed-loop transfer function, T(s), of the system.

ii. Hence, analyze T(s) to **calculate** the range of K for stability.

b. **Find** T(s) and use Routh Hurwitz criteria to **calculate** K that will place the closed-loop poles of the system as shown in the Figure; one pole on real-axis and two poles on $j\omega$-axis. The open loop transfer function G(s) of unity feedback system is expressed below. Hence, **find** the value of the poles as well.



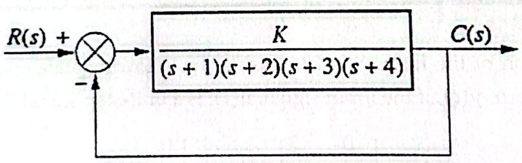
$$G(s) = \frac{Ks^2(s+1)}{s^3+s^2+2s+K}$$

$1 + G(s) = 0$

CLO # 4: Analyze a given system in frequency domain.

Q4: **[20 marks]**

a) For the system shown in Figure below,



i. **Find** the asymptotes (if any) of the root-locus.

ii. **Find** the break-away and break-in points (if any) of the root-locus. $\frac{1}{\sigma - p} = \frac{1}{\sigma - z}$

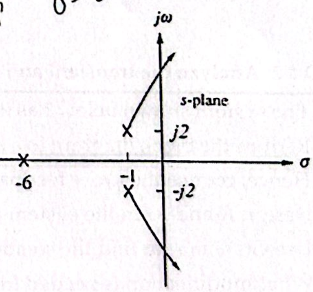
iii. **Find** the crossings on the $j\omega$-axis of the root-locus. $1+G(s)$

iv. Hence, **illustrate** the diagram of the root-locus.

b) The root locus of a system, G(s), is shown here:

i. **Find** the open-loop transfer function, G(s), of the system.

ii. Hence, using the appropriate property, **find** the gain K for which the system will yield a closed-loop pole on the real-axis at **s = −10**.



$1 + G(s) = 0$
$s = -10$

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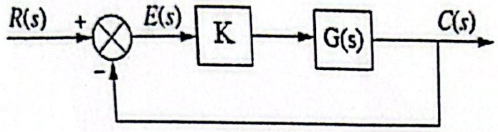
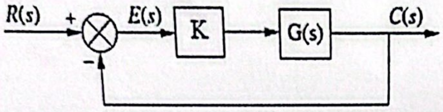


CLO # 5: Design a closed-loop feedback controller to meet transient and steady-state performance specifications.

Q5: **[20 marks]**

The open loop transfer function of an un-compensated unity feedback system is given below.

$$K\,G(s) = \frac{K}{(s+1)(s+5)}$$



The dominant pole of the un-compensated system occurs at $K = 25$.

a. **Find** the location of the dominant pole of the un-compensated system.

b. **Find** the un-compensated system's settling-time, peak-time, and percentage overshoot.

c. It is desired to compensate the system to <u>decrease its settling-time by a factor of 2.0 without affecting its overshoot</u>. **Find** the new settling-time, peak-time, and the new dominant pole location to satisfy the required performance metrics.

d. **Design** a PD controller to satisfy the desired performance metrics. Write the transfer function of the controller including its pole/zeroes.

e. **Construct** an op-amp circuit for the PD controller designed in (d). Usa a capacitor of 10 μF.